MANUAL AIR CONDITIONER SYSTEM [MANUAL AIR CONDITIONER]

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Outline

- The manual air conditioner system is controlled by the instrument cluster.
- The instrument cluster performs the following controls based on the input signals from the each dial/switch and each sensor of the climate control unit.

Basic control	Control description
Air intake control (RECIRCULATE/FRESH switch type climate control unit)	• Air intake actuator motor protection function • FRESH control in defroster mode • RECIRCULATE mode control under severe heat • MAX A/C control (With MAX A/C control)
A/C compressor control	• A/C signal (magnetic clutch) ON/OFF determination • A/C cut-off control • MAX A/C control (With MAX A/C control)

Supplementary function
Air conditioner i-stop control (With i-stop)
Fail-safe

Function

Air intake control (RECIRCULATE/FRESH switch type climate control unit)

- The climate control unit switches the air intake mode by the operation of the RECIRCULATE switch or FRESH switch.

Air intake mode	RECIRCULATE indicator light	FRESH indicator light
FRESH → RECIRCULATE	On	Off
RECIRCULATE → FRESH	Off	On

Air intake actuator motor protection function

- For the air intake actuator motor protection function, if the climate control unit detects an excessive load on the motor while the air intake actuator is operating, it stops the air intake actuator motor.

FRESH control in defroster mode

- The FRESH control in the defroster mode switches the intake mode to FRESH when the airflow mode is set to the defroster.
- The FRESH control in the defroster mode improves the effect of defrosting by switching the air intake mode to FRESH.

RECIRCULATE mode control under severe heat

- When the vehicle is driven at a very low speed with the A/C on under severe heat conditions, the climate control unit may change the air intake mode from FRESH to RECIRCULATE automatically. Due to this, load on the A/C compressor is reduced.

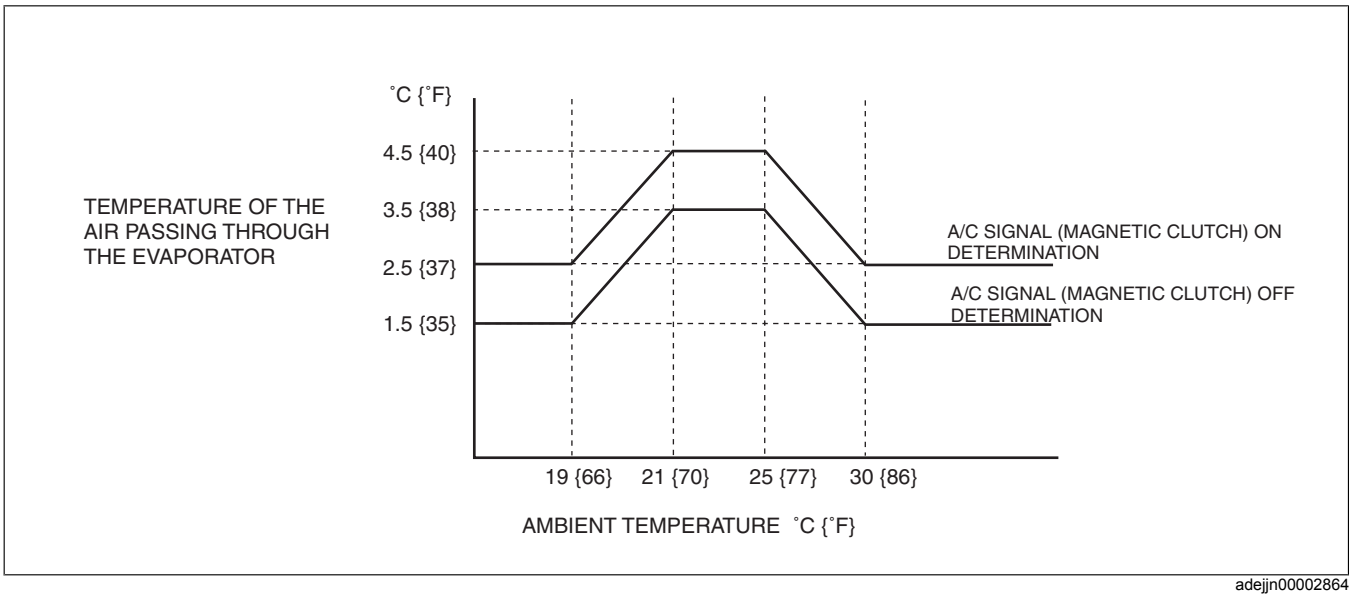
MAX A/C control (With MAX A/C control)

- When the climate control unit detects a temperature setting dial MAX COLD position signal, it switches the air intake mode to RECIRCULATE.

A/C compressor control

A/C signal (magnetic clutch) ON/OFF determination (With heater system)

- When the instrument cluster detects that the airflow volume control dial and A/C switch are turned on, it performs on or off determination (magnetic clutch on or off) based on the ambient temperature and the airflow temperature passing through the evaporator.

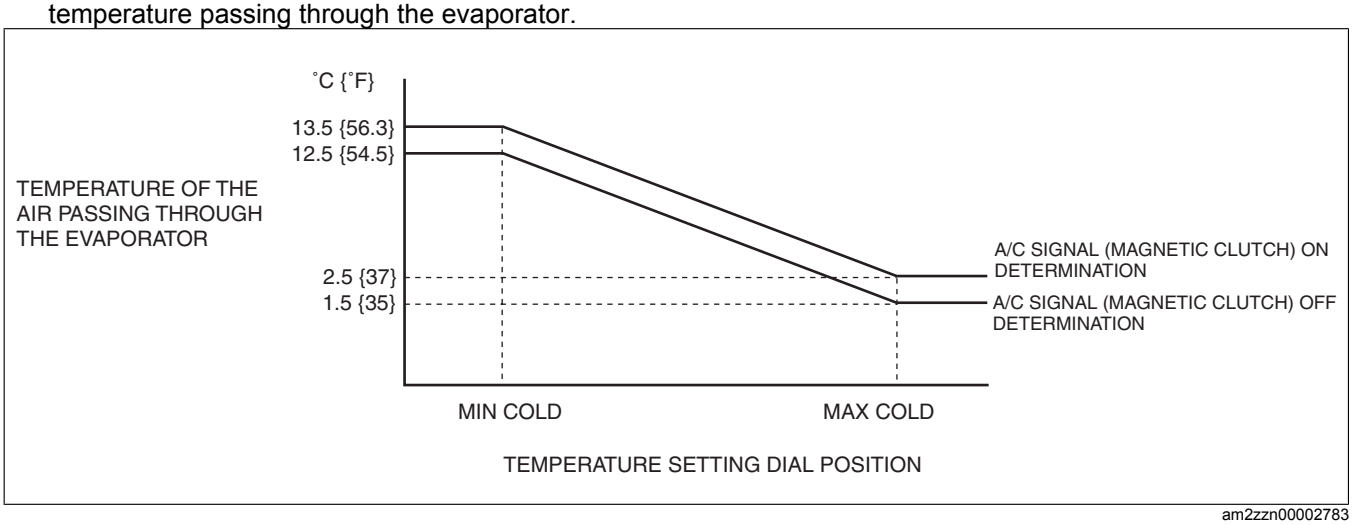


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- The instrument cluster performs control so that the surface temperature of the evaporator remains within a constant range when the airflow volume control dial and A/C switch are on to prevent the evaporator from freezing.
- When the ambient temperature is low, the instrument cluster sets the A/C signal (magnetic clutch) off-temperature to low to prevent window fogging.

A/C signal (magnetic clutch) ON/OFF determination (Without heater system)

- When the instrument cluster detects that the airflow volume control dial and A/C switch are turned on, it performs on or off determination (magnetic clutch on or off) based on the temperature setting dial position and the airflow temperature passing through the evaporator.



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- The instrument cluster performs control so that the surface temperature of the evaporator remains within a constant range when the airflow volume control dial and A/C switch are on to prevent the evaporator from freezing.
- When the set temperature is low, the instrument cluster sets the A/C signal (magnetic clutch) off-temperature to low to prevent window fogging.

A/C cut-off control

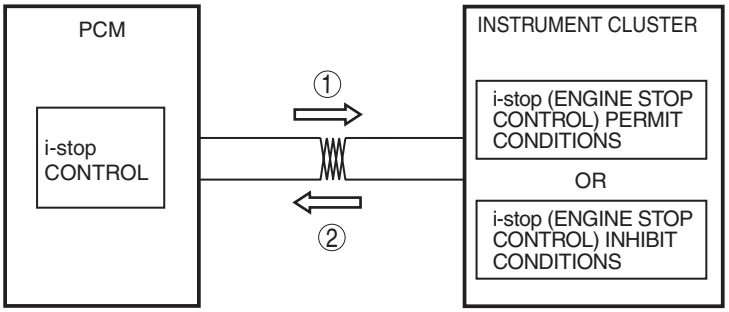
- The PCM performs A/C cut control based on the input signal from the instrument cluster and outputs an operation signal to the A/C relay (magnetic clutch). (See A/C CUT-OFF CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)(See A/C CUT-OFF CONTROL [SKYACTIV-D 1.5].)

MAX A/C control (With MAX A/C control)

- When the instrument cluster detects a temperature setting dial MAX COLD position signal from the climate control unit, it determines that the A/C signal (magnetic clutch) is on.

Air conditioner i-stop control (With i-stop)

- The instrument cluster determines i-stop control (engine stop control) according to the operation condition of the air conditioner system and sends i-stop (engine stop control) request signals to the PCM.



① : i-stop STATUS SIGNAL

② : i-stop REQUEST SIGNAL
(i-stop (ENGINE STOP CONTROL) PERMIT /
i-stop (ENGINE STOP CONTROL) INHIBIT)

: HS-CAN

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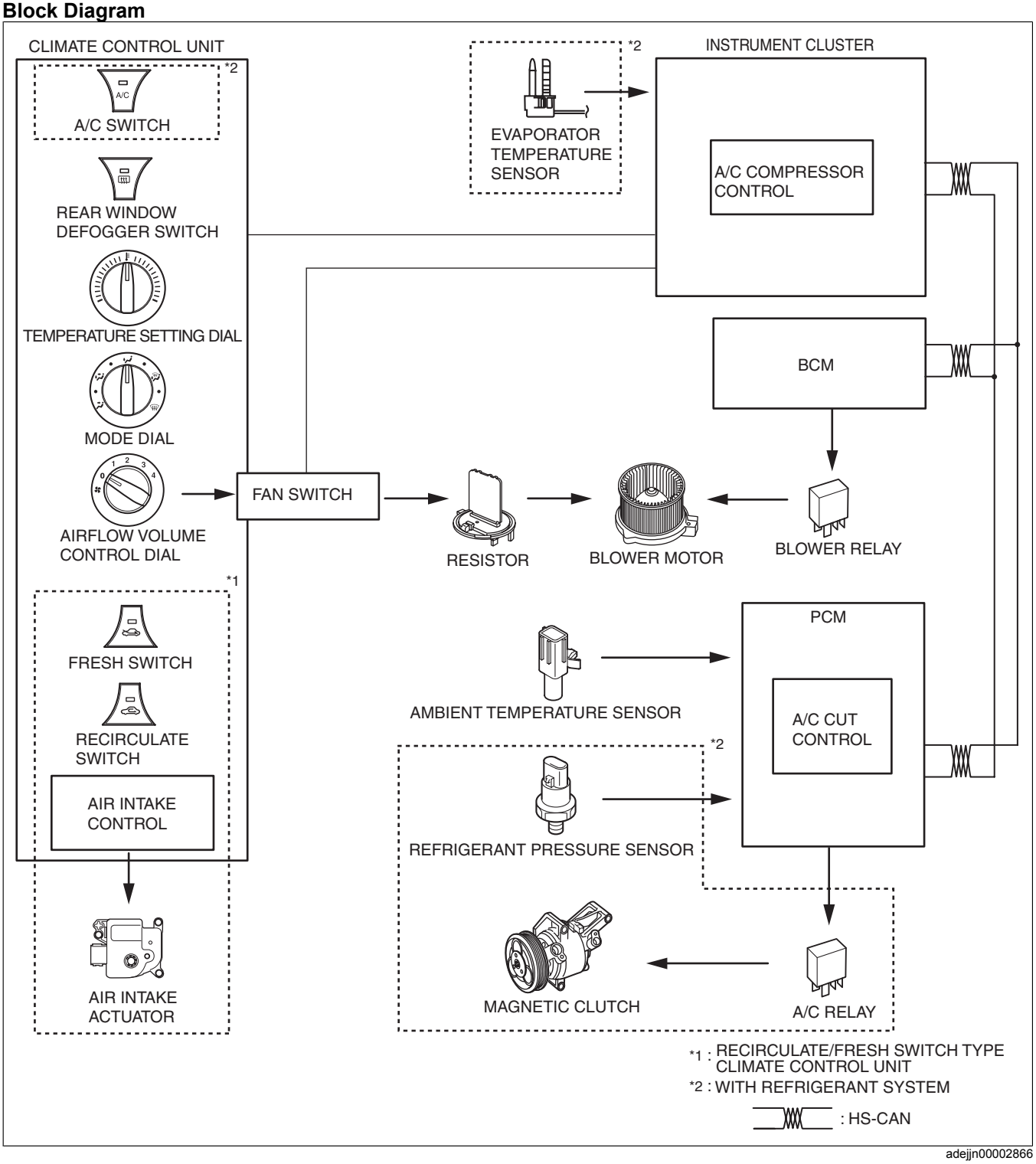
Fail-safe

- If a malfunction is detected by the malfunction detection function, the fail-safe function performs the following controls to prevent an operation air conditioner malfunction and malfunction of output devices.

Fail-safe function table

Part where malfunction is determined	Malfunction determined when ignition is switched ON	Ignition switched ON under malfunction condition
Ambient temperature sensor	Ambient temperature sensor input value is fixed at value right before malfunction	Ambient temperature sensor input value fixed at 15 °C {59 °F}
Evaporator temperature sensor	Evaporator temperature sensor input value fixed at **0 °C {32 °F}**	←
Engine coolant temperature sensor	Engine coolant temperature sensor input value fixed at **85 °C {185 °F}**	←

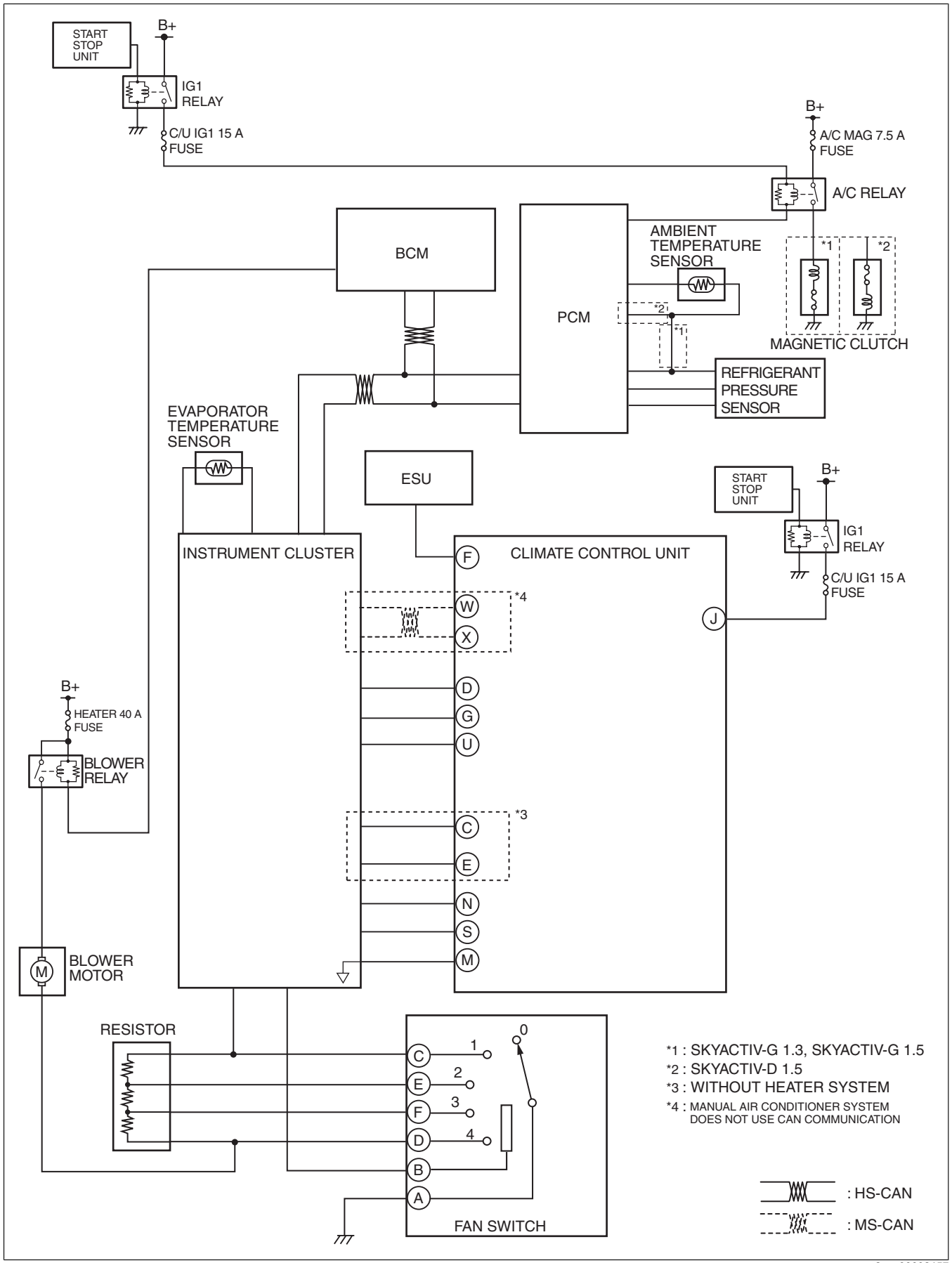
Block Diagram



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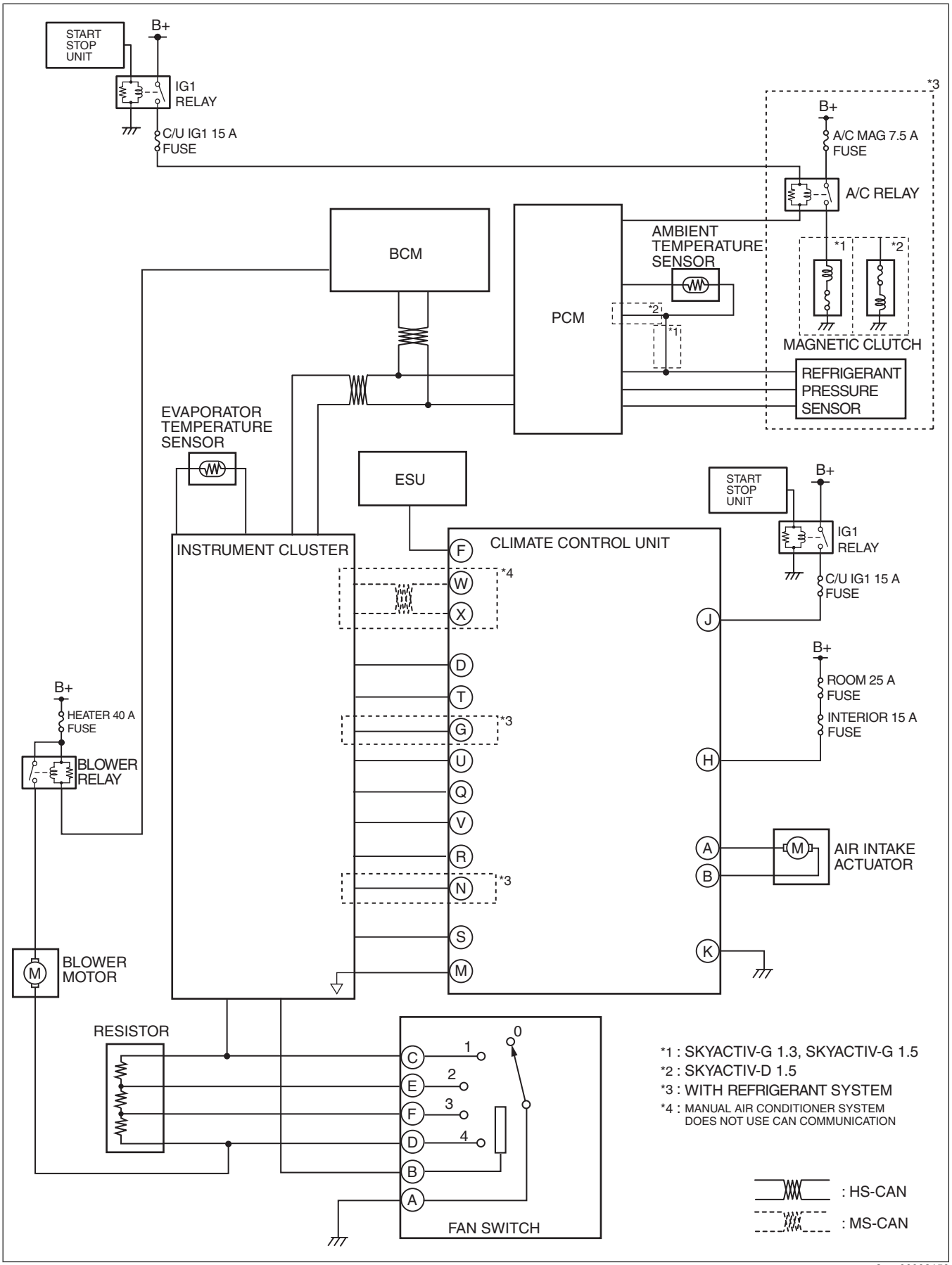
System Wiring Diagram

RECIRCULATE/FRESH lever type climate control unit



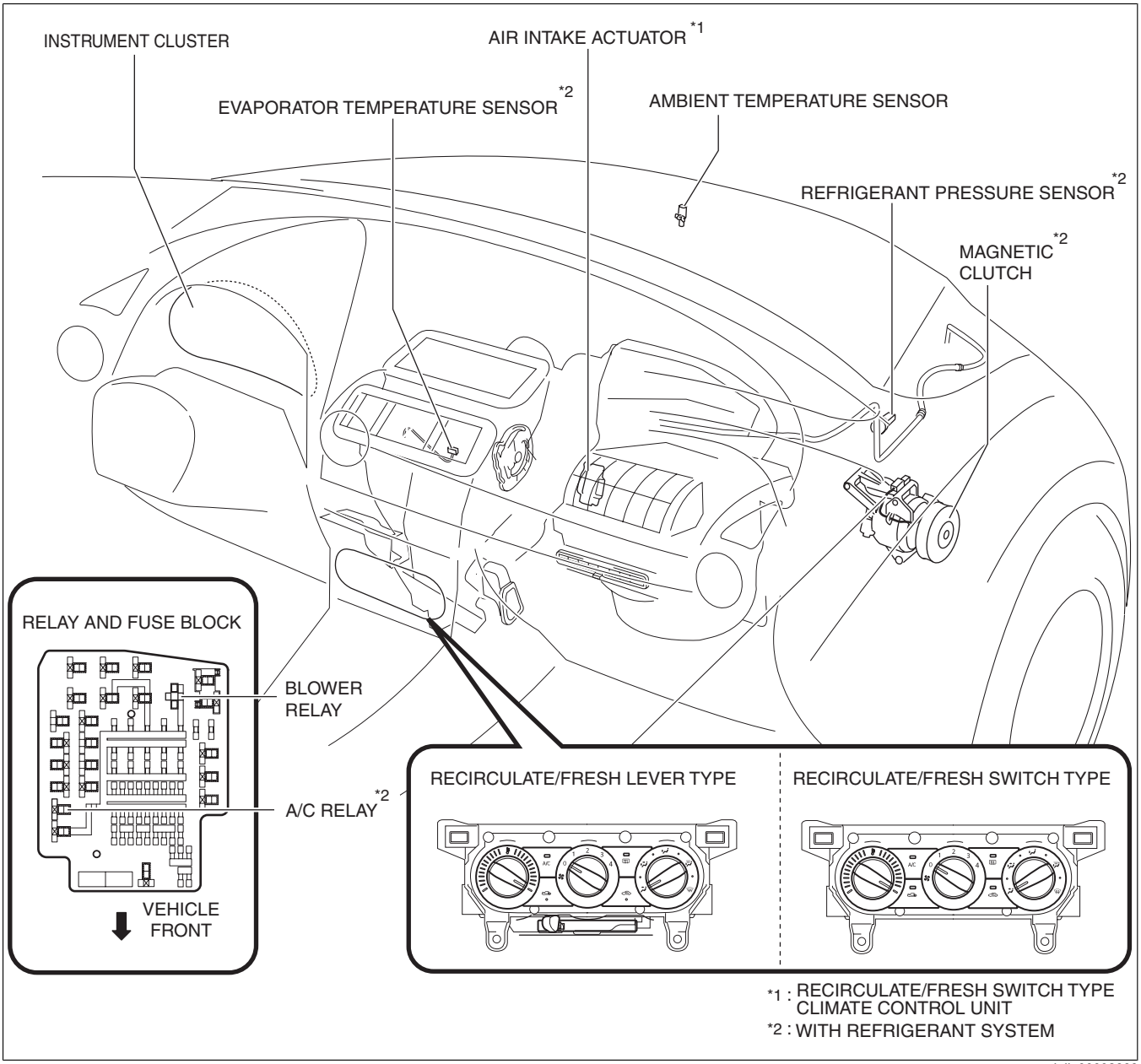
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RECIRCULATE/FRESH switch type climate control unit



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Structural View

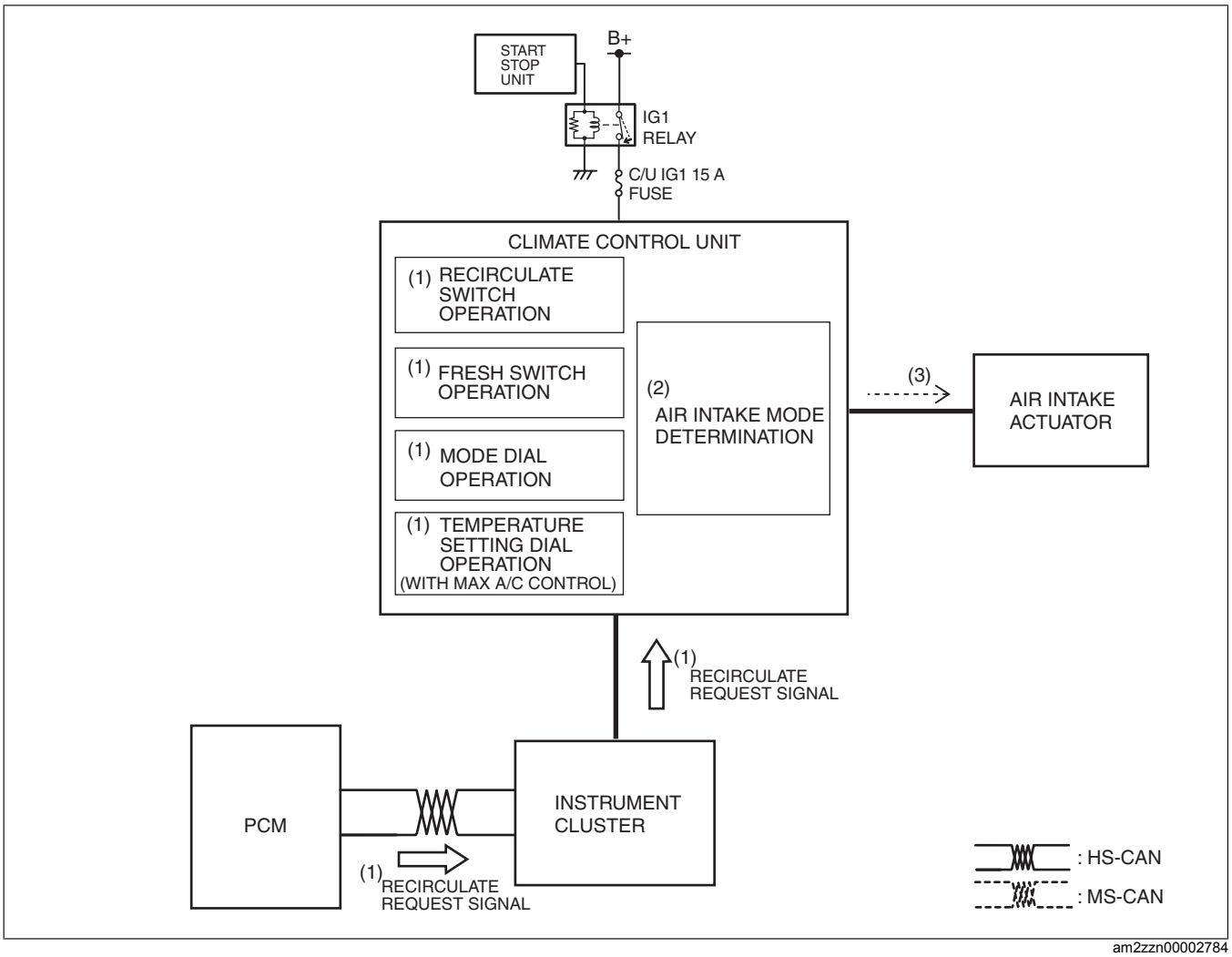


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Operation

Air intake control (RECIRCULATE/FRESH switch type climate control unit)

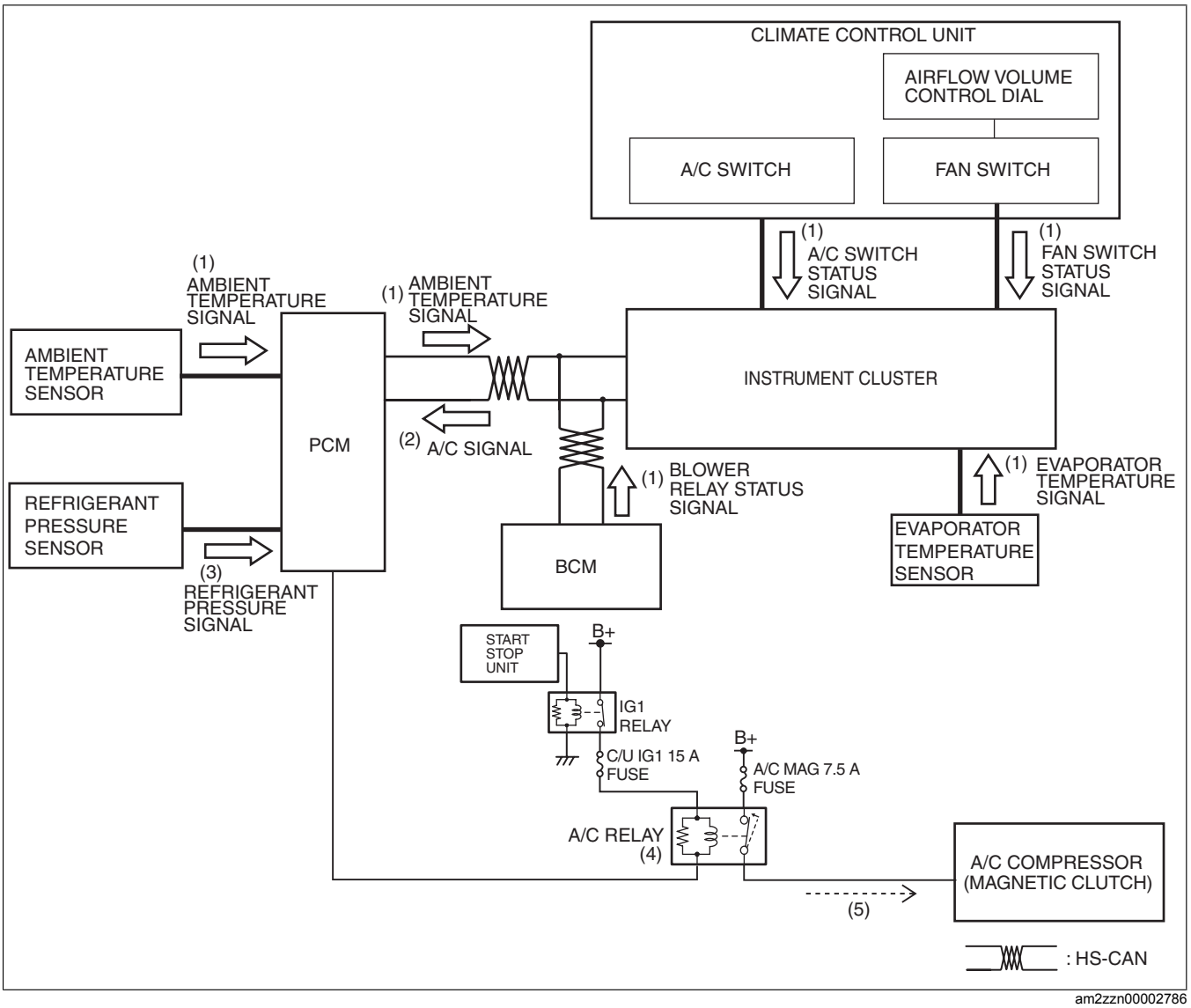
1. The climate control unit performs air intake mode determination (2) based on the signals (1) from each sensor which change according to the operation of the RECIRCULATE switch and the FRESH switch and the vehicle conditions when the ignition is switched ON (engine off or on).
2. The climate control unit drives (3) the air intake actuator for a specified time based on the results of the air intake mode determination and corrections.



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A/C compressor control (With heater system)

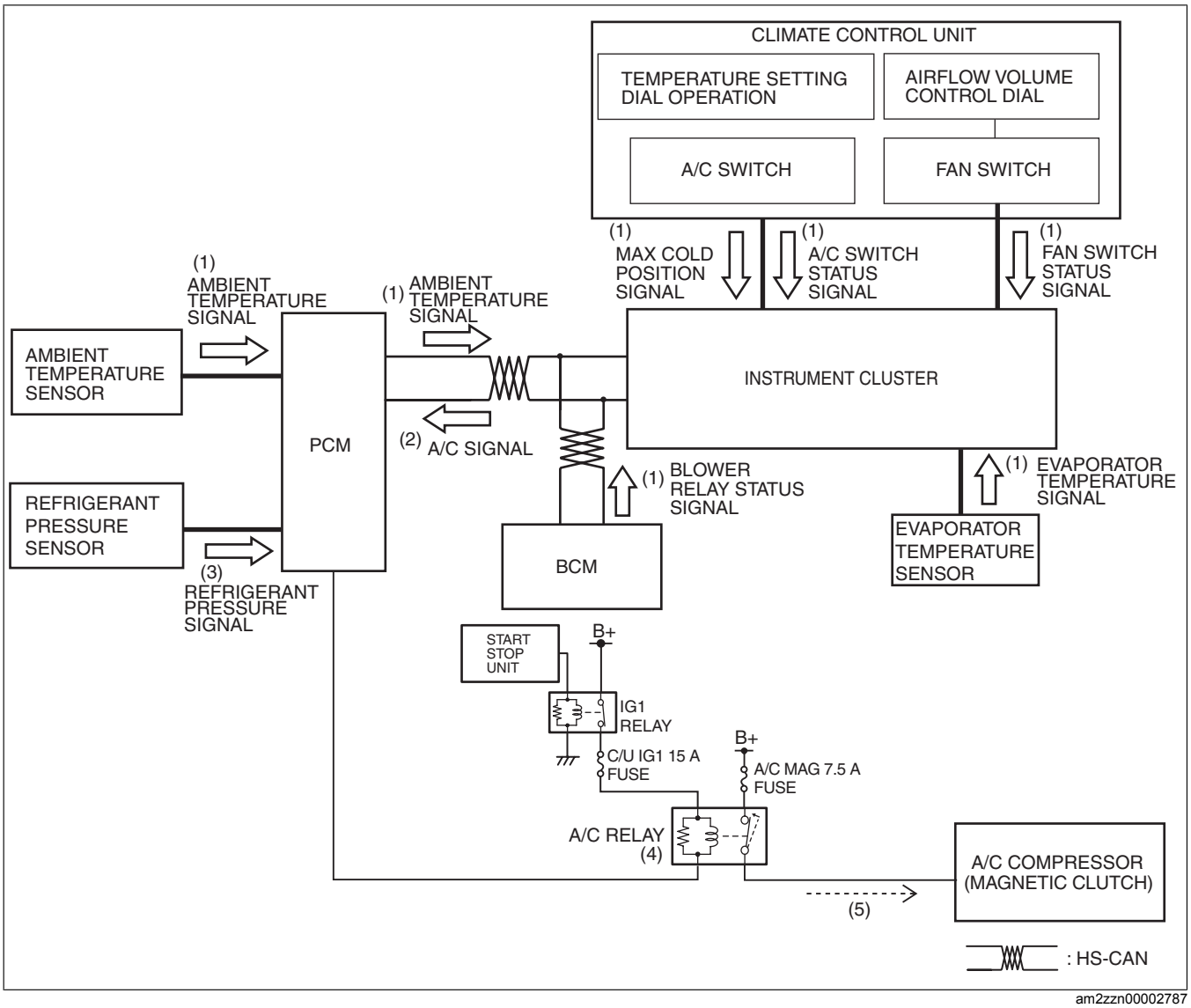
1. The instrument cluster sends an A/C signal (2) to the PCM based on the signals (1) from each sensor which change according to the climate control unit A/C switch, airflow volume control dial, and the vehicle conditions.
2. The PCM turns the A/C relay on (4) based on the signals (3) from each sensor which change according to the the A/C signal and the vehicle conditions.
3. When the A/C relay turns on, the magnetic clutch turns on (5).



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A/C compressor control (Without heater system)

1. The instrument cluster sends an A/C signal (2) to the PCM based on the signals (1) from each sensor which change according to the climate control unit A/C switch, temperature setting dial, airflow volume control dial, and the vehicle conditions.
2. The PCM turns the A/C relay on (4) based on the signals (3) from each sensor which change according to the the A/C signal and the vehicle conditions.
3. When the A/C relay turns on, the magnetic clutch turns on (5).



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Air conditioner i-stop control (With i-stop)

i-stop request control (i-stop (engine stop control) permission)

- The instrument cluster sends an i-stop (engine stop control) permission request signal to the PCM in conjunction with the operation conditions of the air conditioner system.

i-stop (engine stop control) permission request

- The instrument cluster sends an i-stop (engine-stop control) permit signal to the PCM when it does not detect any of the following conditions:

i-stop (engine stop control) inhibit conditions

No.	Item	Vehicle condition
1	Blower motor control status	• Fan switch in on (1st-4th) • And during defroster control
2	Ambient temperature	Ambient temperature is **-10 °C {14 °F} or less**, or **50 °C {122 °F} or more**
3	Engine coolant temperature	When the fan switch is on (1st-4th), the engine coolant temperature does not meet the i-stop permit conditions (comfortable cabin temperature control not performed)
4	Evaporator temperature	When the A/C switch is on, the evaporator temperature does not meet the i-stop permit conditions (comfortable cabin temperature control not performed)